MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Course Plan

Course Name	:	Reinforcement Learning
Course Code	:	AME 5232
Academic Year	:	2025-2026
Semester	:	II
Name of the Course Coordinator	:	Dr. Devi Prasad M
Name of the Program Coordinator	:	Mr. Sudarsan N S Acharya

Date: 7/1/2026	
Signature of Program Coordinator	
with Date	

Date: 7/1/2026	
Signature of Course Coordinator	
with Date	

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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning), programare as follows

PEO No.	Education Objective
PEO 1	Acquire solid mathematical and computational skills essential for understanding, applying, and developing modern artificial intelligence and machine learning algorithms.
PEO 2	Produce industry-ready graduates with practical experience in structuring machine learning projects using state-of-the-art software.
PEO 3	Address multi-disciplinary challenges through coursework and projects by adapting to the rapidly advancing developments in artificial intelligence and machine learning approaches.

Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning), graduates will be able to:

PO 1	An ability to independently carry out research/investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	Demonstrate a degree of mastery over the area as per the specialization of the program which should be at a level higher than the requirements in an appropriate bachelor program.
PO 4	Identify appropriate and efficient algorithmic approaches for solving real-life challenges using artificial intelligence and machine learning principles and state-of-the art software prevalent in industry and academia.
PO 5	Develop teamwork and leadership skills for addressing challenges of social importance for sustainable societal development using ethical artificial intelligence and machine learning approaches.

1. Course Plan

1.1 Primary Information

Course Name	:	Reinforcement Learning [AME 5232]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Probability Theory
Core/ PE/OE	:	PE

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No. of Contact Hours	Program Outcomes (POs)	BL
CO1	Distinguish reinforcement learning from non-interactive machine learning.	10	PO4	4
CO2	Formulate an application problem as a reinforcement learning problem.	16	PO4	6

CO3	Apply state of the art reinforcement learning algorithms.	10	PO4	3

1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	15 days	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply, Analyze, Create	Apply, Create	Apply, Analyze, Create
Pattern	Answer all five questions of 10 marks each	Answer all questions	Answer all ten questions of 10 marks each
Schedule	As per academic calendar.	February 2026 and March 2026	As per academic calendar.

Topics covered	Introduction. Markov decision processes. Bandit problems. Agent-Environment interface. Goals, rewards, and returns. Value functions.	Representing Markov chains. Markov processes. Gymnasium usage.	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.

1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Examples and elements of reinforcement learning	CO1
L2	Limitations and scope of reinforcement learning - History of reinforcement learning	CO1
L3	N-Armed bandit problem: action-value methods - 1	CO2
L4	N-Armed bandit problem: action-value methods - 2	CO2
L5	N-Armed bandit problem: action-value methods - 3	CO2
L6	Finite Markov Decision Process - 1	CO2
L7	Finite Markov Decision Process - 2	CO2

L8	The agent–environment interface - 1	CO1
L9	The agent–environment interface - 2	CO1
L10	Goals and rewards, returns - 1	CO1
L11	Goals and rewards, returns - 2	CO1
L12	Markov decision processes - 1	CO1
L13	Markov decision processes - 2	CO3
L14	Value functions, and optimal value functions - 1	CO3
L15	Value functions, and optimal value functions - 2	CO3
MSE	Mid Semester Exam	CO1, CO2, & CO3
L16	Generalized policy iteration	CO3
L17	Generalized policy iteration	CO3
L18	Importance of exploration	CO2
L19	Importance of exploration	CO3
L20	Monte Carlo Prediction	CO1

L21	Monte Carlo estimation of action values	CO2
L22	Monte Carlo control without exploring starts	CO2
L23	Temporal difference methods for control - TD Prediction	CO2
L24	Temporal difference methods for control - Optimality of TD(1)	CO2
L25	Temporal difference methods for control - SARSA	CO2
L26	Temporal difference methods for control - SARSA	CO2
L27	Q-Laerning - Off-policy TD Control	CO1
L28	Maximization Bias and Double Learning	CO3
L29	Games, Afterstates and other special cases	CO3
L30	Models of planning	CO2
L31	Integrated planning, acting and learning	CO2
L32	Integrated planning, acting and learning	CO3
L33	Expected versus sample updates	CO1
L34	Real-time dynamic programming	CO2

L35	Planning at decision time, Rollout Algoirthms, Monte Carlo Tree Search	CO2
L36	Integrating supervised & reinforcement learning.	CO3
ESE	End Semester Exam	0

1.5 References

1. Mathematical Foundations of Reinforcement Learning, Shiyu Zhao; Springer, 2025.
2. Deep Reinforcement Learning Hands-On, Maxim Lapan, 2nd Edition; Packt, 2020.
3. Reinforcement Learning: An Introduction, Richard S. Sutton and Andrew G. Barto, MIT Press, 2nd Edition – Available online at https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf
4. Fundamentals of Reinforcement Learning (University of Alberta, https://www.coursera.org/learn/fundamentals-of-reinforcement-learning)

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.

1.7 Course Timetable

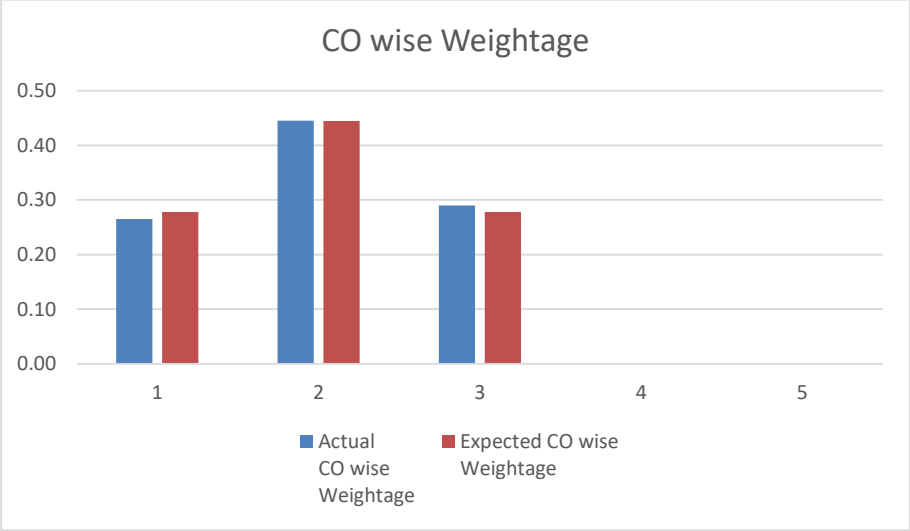
II Semester						Room: LG01-LH12			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday			RL	RL					
Tuesday									
Wednesday									
Thursday									
Friday									

Saturday		RL	

1.8 Assessment Plan

COs		Marks & Weightage				
CO No.	**CO Name**	**MSE Marks (Max. 50)**	**Assignment Marks (Max. 20)**	**ESE Marks (Max. 100)**	**Actual CO wise Weightage**	**Expected CO wise Weightage**
CO1	Distinguish reinforcement learning from non-interactive machine learning.	15		35	**0.27**	**0.28**
CO2	Formulate an application problem as a reinforcement learning problem.	20	15	35	**0.45**	**0.44**
CO3	Apply state of the art reinforcement learning algorithms.	15	5	30	**0.29**	**0.28**

	Marks (weightage)	**0.3**	**0.2**	**0.5**	**1.00**	**1.00**



Note:
• In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
• ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
• ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

Weightage for CO1 = (MSE marks for CO1 / 1.6666 + Assignment marks for CO1/1.0 + ESE marks for CO1 / 2)/100

1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

SI. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	• Performance is measured using MSE attainment level. • Reference: question paper and answer scheme. • MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	2	• Performance is measured using assignments/quiz attainment level. • Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	• Performance is measured using ESE attainment level. • Reference: question paper and answer scheme. • ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1				Y	
CO2				Y	
CO3				Y	
Average Articulation Level				*	